# Internal Structure of the Equilateral Lattice Path Integral: Crystal Angular Momentum, Emergent Vector Ground State, and Symmetry Breaking by Confinement

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In a previous paper [1] we showed that an equilateral triangular (1+1D) and hexagonal (2+1D) path-integral lattice with the single amplitude rule "direction change $= i\varepsilon$" reproduces relativistic dispersion, exact isotropy, and the speed of light $c = \sqrt{3}$ from geometry alone. Here we analyse the *internal* structure of the physical propagating band of the same model, in 2+1D and in a natural 3+1D tetrahedral-octahedral extension. In 2+1D the physical band is exactly 5-fold degenerate and decomposes under $C_{6v}$ as $E_1 \oplus E_2 \oplus B$, carrying crystal angular momenta $m \in \{\pm 1, \pm 2, +3\}$ modulo 6; the $s$-wave is excluded by a factor-6 spectral gap in the local coupling matrix. In 3+1D the band is exactly 11-fold degenerate and decomposes under $O_h$ as $E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}$; the exclusion factor is 12. A general law follows: for $N$ lightlike diagonal directions the band has dimension $N-1$ and the $s$-wave eigenvalue of the coupling matrix is $1 + Ni\varepsilon$. A numerical double-slit experiment shows that a free wave packet is 98% $T_{1u}$ (vector). A 3+1D box-state calculation with hard walls places $T_{1u}$ consistently lowest in energy, with the $T_{1u}$–$E_g$ gap following $1/L^2$ to within 2%. Vector physics emerges geometrically as the preferred ground state of the lattice.

## I. INTRODUCTION

In Ref. [1] we introduced an equilateral path-integral lattice in which every edge has length one, a single amplitude rule assigns a factor $i\varepsilon$ to every direction change and 1 to every straight continuation, and all relativistic structure emerges directly from the geometry. The 1+1D equilateral triangular version gives a speed of light $c = \sqrt{3}$, a physical mass $m_{\text{phys}} = \arctan\bigl(2\varepsilon/(1-\varepsilon^2)\bigr) \approx 2\varepsilon$, and the correct relativistic time dilation in the narrow-packet limit. The 2+1D hexagonal extension inherits these properties with *exact* 6-fold isotropy and no fermion doubling.

One feature of the 2+1D analysis was left open in Ref. [1]: the physical propagating band at $k = 0$ is exactly 5-fold degenerate, but its representation-theoretic content and physical meaning were not identified. This paper answers that question and extends the analysis to a 3+1D face-centred-cubic (cuboctahedron) lattice.

Three results emerge. First, the physical band is always the *traceless part of the direction-space Fourier representation*: its dimension is one less than the number $N$ of lightlike diagonal directions, and the excluded state is the $s$-wave. The mechanism is a spectral gap in the local coupling matrix $C$: the $s$-wave gets eigenvalue $1 + Ni\varepsilon$, every orthogonal Fourier mode gets $1 - i\varepsilon$, and the factor-$N$ imbalance banishes the $s$-wave to a different band. Second, in 3+1D the physical band decomposes under the full octahedral group $O_h$ as $E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}$, and the "vector" irrep $T_{1u}$ dominates free propagation (98% of a wave packet) and sits lowest in energy in a hard-wall box. Third, higher angular-momentum content (in particular the $d$-wave $m = \pm 2$ in 2+1D) is only excited at obstructions such as slit edges, where it can reach ∼47% of the local amplitude.

Phrased physically, the equilateral lattice geometrically prefers a vector field as its low-energy sector. This is suggestive in view of the fact that every fundamental interaction in the Standard Model—photon, $W^{\pm}$, $Z$, gluons—is mediated by a spin-1 vector boson.

## II. CRYSTAL ANGULAR MOMENTUM IN 2+1D

### A. Symmetry group $C_{6v}$

The full one-step transfer matrix $M_{\text{full}}(k) = M_{\text{half}}^2$ of the 2+1D hexagonal lattice acts on a 14-dimensional state (seven direction amplitudes for the current half-step and seven for the previous). At $k = 0$ it is exactly invariant under the crystallographic point group $C_{6v}$ (order 12, the symmetry group of the regular hexagon). The invariance is verified numerically [1]:

$$\max_{g \in C_{6v}} \bigl\| \, [M_{\text{full}}, R(g)] \, \bigr\| \; < \; 10^{-10}. \tag{1}$$

Here $R(g)$ is the 14×14 joint permutation representation of $g$ on the direction index (last index 6, the straight move, is fixed; the six diagonals cycle).

### B. Decomposition of the physical band

The 14-dimensional state space decomposes into $C_{6v}$ irreps as

$$4A_1 \oplus 2B_1 \oplus 2E_1 \oplus 2E_2,$$

and the 5-fold-degenerate physical band at $E = m_{\text{phys}} = \arctan\bigl(2\varepsilon/(1-\varepsilon^2)\bigr)$ carries exactly the traceless part

$$\text{band}^{(2+1\text{D})} \; = \; E_1 \oplus E_2 \oplus B_1, \qquad \dim = 2 + 2 + 1 = 5. \tag{2}$$

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All five eigenvectors share the eigenvalue $\lambda = (1-\varepsilon^2) - 2i\varepsilon$, giving $|\lambda| = 1+\varepsilon^2$ and $E = m_{\rm phys}$. Joint diagonalisation of $M_{\rm full}$ and the sixfold rotation $R_{60}$ labels the five modes by crystal angular momentum

$$m \in \{-2,\, -1,\, +1,\, +2,\, +3\} \pmod 6, \tag{3}$$

so the $s$-wave $m = 0$ (the $A_1$ irrep) is absent. Figure 1 shows the polar structure of the five modes: each amplitude lives on the six diagonal directions with phase cycling at $m \cdot 60°$, while the straight (timelike) component vanishes identically to machine precision.

### C. The exclusion mechanism

The 7×7 amplitude matrix $C_{dd'}$ (with $C_{dd} = 1$ and $C_{dd'} = i\varepsilon$ for $d \neq d'$) has only two distinct eigenvalues,

$$\lambda_C^{(s\text{-wave})} \;=\; 1 + 6i\varepsilon, \qquad \lambda_C^{(m\neq 0)} \;=\; 1 - i\varepsilon, \tag{4}$$

where the $s$-wave (all-ones) eigenvector constructively sums the six off-diagonal $i\varepsilon$ contributions and every orthogonal Fourier mode destructively sums them to $-i\varepsilon$. The factor-6 mismatch of imaginary parts is exactly the spectral gap that places the $s$-wave in a different band. Explicit projection of $M_{\rm full}(k=0)$ onto each angular-momentum sector of $R_{60}$ confirms this: every $m \neq 0$ sector produces the eigenvalue $E = m_{\rm phys}$, while the $m = 0$ sector produces $E \in \{-0.996, +0.108, +0.007\}$, none of which is in the physical band [1]. Physically: *the Dirac-like dynamics lives in the traceless part of the direction-space Fourier representation.*

## III. EXTENSION TO 3+1D: FULL OCTAHEDRAL SYMMETRY

### A. Lattice geometry

The 3+1D analogue uses the twelve cuboctahedron vertices as lightlike diagonals plus one timelike (straight) direction, giving 13 moves per half-step. The twelve diagonals are the vectors $(\pm a, \pm a, 0)$ and the two cyclic permutations with $a = \sqrt{3}/(2\sqrt{2})$, and the straight move advances time by one. Every edge has the same spacetime length 1, and the speed of light is exact and geometric,

$$c \;=\; \frac{\sqrt{3}/2}{1/2} \;=\; \sqrt{3}. \tag{5}$$

The full one-step transfer matrix is $26 \times 26$ (13 current-direction amplitudes plus 13 previous-direction amplitudes).

### B. $O_h$ invariance

The group $O_h$ (order 48, full octahedral) is realised on the direction index as the 48 signed permutations of the three spatial axes. Numerical verification gives

$$\max_{g \in O_h} \big\| \, [M_{\rm full}(k=0), R(g)] \, \big\| \;=\; 3.14 \times 10^{-16}, \tag{6}$$

i.e. machine zero.

The character of the 11-dimensional physical band on the ten conjugacy classes of $O_h$ is

$$\chi_{\rm band} \;=\; (11,\, -1,\, -1,\, -1,\, +1,\, -1,\, -1,\, -1,\, +3,\, +1), \tag{7}$$

in the class order ($E$, $8C_3$, $3C_2$, $6C_4$, $6C_2'$, $i$, $6S_4$, $8S_6$, $3\sigma_h$, $6\sigma_d$). Applying character orthogonality to the standard $O_h$ table yields integer multiplicities.

### C. Decomposition

The full 13-dimensional direction-space representation decomposes as

$$\text{dir}^{(3+1\text{D})} \;=\; 2A_{1g} \oplus E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}, \tag{8}$$

and the physical band at $E = m_{\rm phys} \approx 0.199$ is

$$\boxed{\text{band}^{(3+1\text{D})} \;=\; E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u},} \tag{9}$$

with dim $= 2+3+3+3 = 11$, matching the observed degeneracy to machine precision (all 11 eigenvectors have residual $\|M\psi - \lambda\psi\| \lesssim 10^{-15}$). The two excluded $A_{1g}$ states are (i) the straight timelike direction, trivially $O_h$-invariant, and (ii) the $s$-wave of the twelve diagonals, the all-ones cuboctahedron mode. Both are removed from the physical band by the factor-12 spectral gap

$$\lambda_C^{(s)} \;=\; 1 + 12i\varepsilon, \qquad \lambda_C^{(m\neq 0)} \;=\; 1 - i\varepsilon. \tag{10}$$

The straight component of every band eigenvector is identically zero to machine precision, as in the lower-dimensional cases: the physical states are pure superpositions of the twelve lightlike diagonals.

### D. The general exclusion law

The mechanism is independent of dimension: for any equilateral lattice whose moves comprise $N$ lightlike diagonal directions plus (optionally) one straight direction, the local coupling matrix $C_{dd'} = \delta_{dd'} + i\varepsilon(1-\delta_{dd'})$ on the $N \times N$ diagonal block has eigenvalues

$$\lambda_C^{(s)} = 1 + Ni\varepsilon, \qquad \lambda_C^{(m\neq 0)} = 1 - i\varepsilon, \tag{11}$$

the first of multiplicity 1 and the second of multiplicity $N-1$. The factor-$N$ gap places the $s$-wave in a different band, and the physical propagating band has dimension $N-1$. Table I summarises the three instances explicitly verified in Ref. [1] and in the present work.

Figure 3 shows the band fanning out along $k_x$: the 11-fold multiplet decomposes into the little-group $C_{4v}$ irreps $A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E$ as soon as $k \neq 0$.

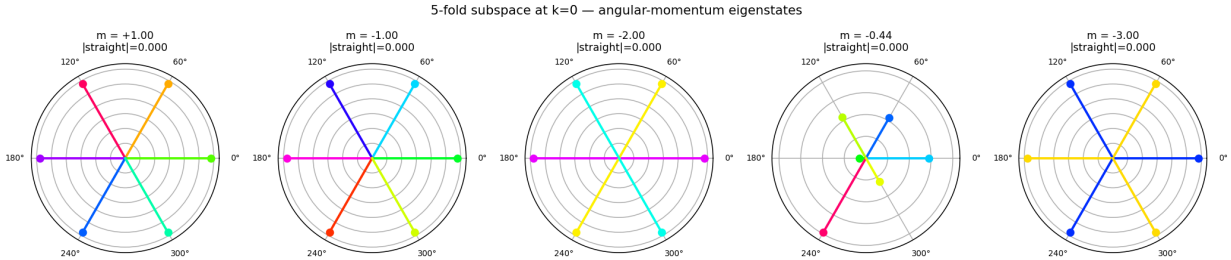


FIG. 1. Polar plots of the five physical eigenvectors of the 2+1D hexagonal transfer matrix at $k = 0$. Each eigenvector is supported on the six lightlike diagonal directions (straight component identically zero) with phase cycling at $m \cdot 60^\circ$ around the hexagon. The five modes carry crystal angular momenta $m \in \{-2, -1, +1, +2, +3\}$ (mod 6); the $s$-wave $m = 0$ is excluded by the spectral gap (4).

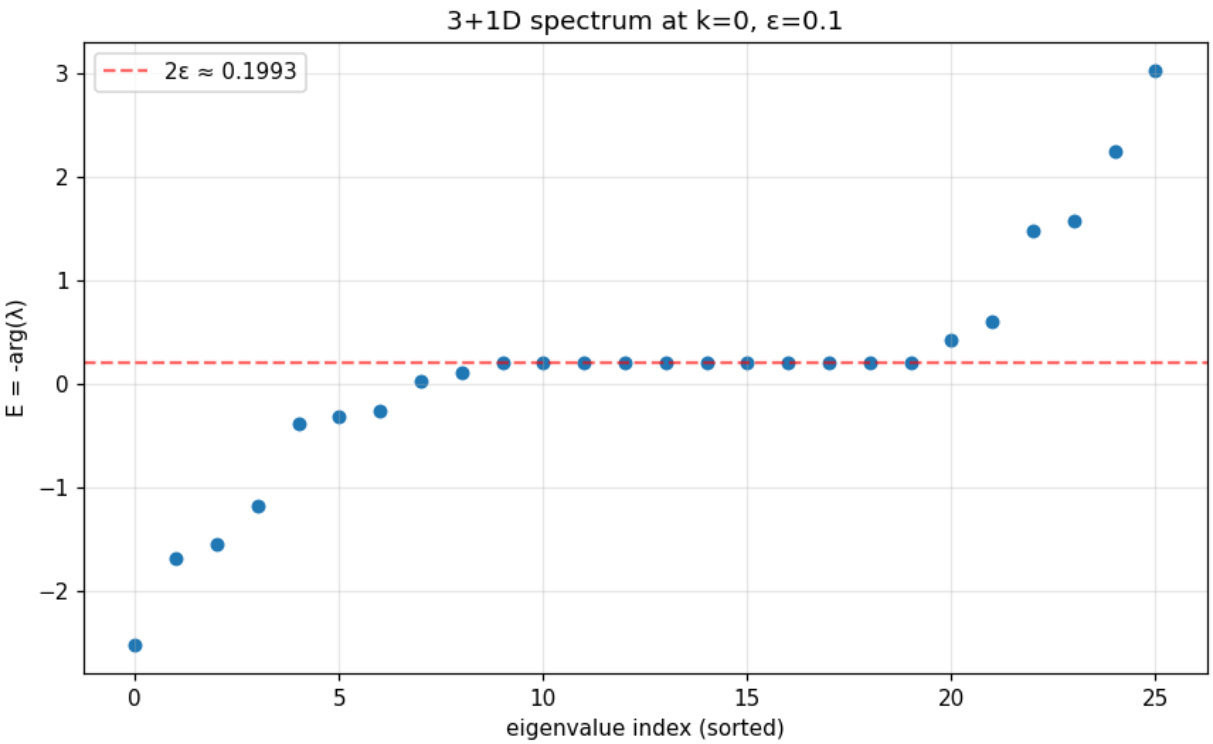


FIG. 2. Eigenvalue spectrum of the 26×26 full-step transfer matrix of the 3+1D tetrahedral-octahedral lattice at $k = 0$ with $\varepsilon = 0.1$. The rank of $M_{\text{full}}$ is 13 (thirteen kernel modes at $|\lambda| = 0$, as in the lower-dimensional cases). Eleven of the thirteen propagating eigenvalues coincide at the physical band $E = m_{\text{phys}} = \arctan\big(2\varepsilon/(1-\varepsilon^2)\big) \approx 0.199$, shown as the dashed line.

## IV. DOUBLE-SLIT EXPERIMENT

### A. Setup

A Gaussian wave packet with width $\sigma = 6$ and momentum $k_0 = 0.3$ is launched from $(x, y) = (8, 60)$ on a 120×120 hexagonal grid. A barrier is placed at $x = 30$ with two slits at $y \in [52, 56]$ and $y \in [64, 68]$; barrier nodes have their amplitude set to zero after every half-step. Time evolution is run for $T = 60$ physical time steps (120 half-steps) and the amplitude at every grid point is projected onto the 5-dimensional physical band of $M_{\text{full}}(k = 0)$ before the density is taken, which suppresses non-physical fast modes. Figure 4 shows snapshots at $t = 20, 40, 60$.

### B. Mode content of the free wave

Each local amplitude is decomposed into the five angular-momentum eigenstates $|\psi_m\rangle$ using joint diagonalisation of $M_{\text{full}}$ and $R_{60}$ within the band. Table II lists

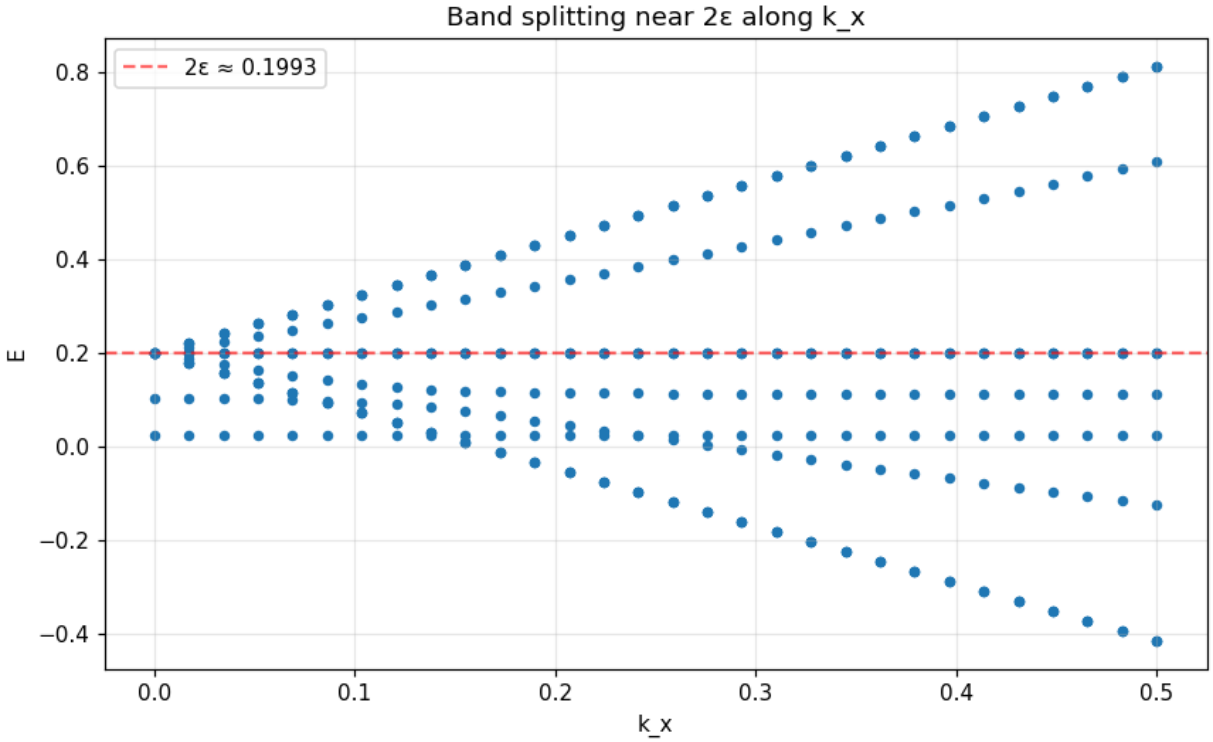


FIG. 3. Splitting of the 11-fold physical band along $k_x$ in the 3+1D cuboctahedral lattice. At $k = 0$ all 11 sub-bands coincide at $E = m_{\rm phys}$ (dashed line); for $k \neq 0$ the little group of $k_x$ is $C_{4v}$ and the $O_h$ multiplet reduces to $A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E$.

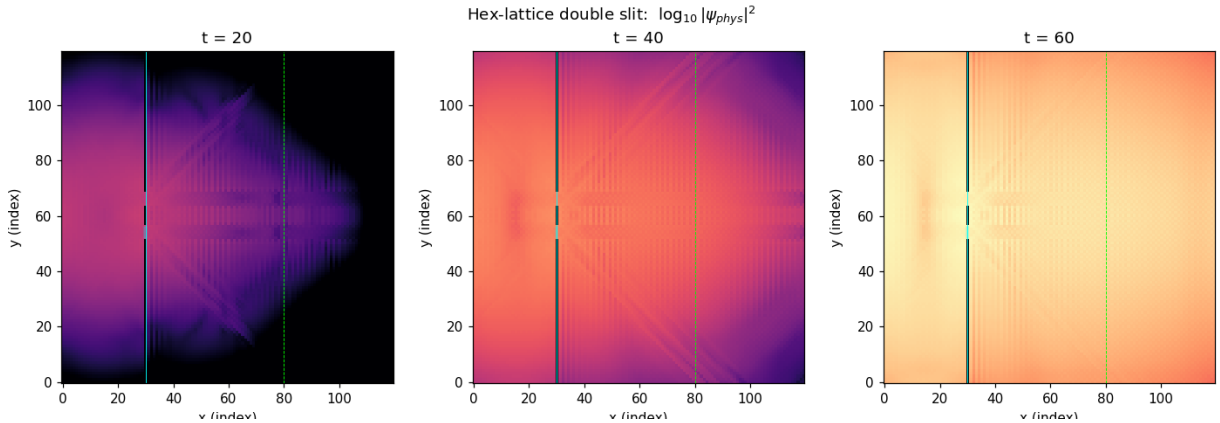


FIG. 4. Physical-band probability density $|\psi_{\rm phys}|^2$ (log scale) on the 2+1D hexagonal lattice double-slit geometry at $t = 20, 40, 60$ half-step periods. Cyan vertical line: barrier with two open slits. Dashed green line: screen readout at $x = 80$.

the fractional $m$-content at several points at $t = 60$.

Two observations follow. First, the free propagating wave is 98.4% in the $m = \pm 1$ sector—the $E_1$ ($T_{1u}$-like) "vector" irrep. This is not put in by hand; any sufficiently smooth wave packet on the hexagonal lattice produces it automatically. Second, the slit *centers* pick one chirality of the vector field (slit A, above the axis, enhances $m = -1$; slit B mirror-reflects), while the slit *edges* enhance $d$-wave content ($m = \pm 2$) by a factor of almost 80, from $\sim$1.2% in the free wave to $\sim$47% at the edge.

### C. Symmetry breaking at the slit

Physically the slit-edge enhancement is a lattice analogue of the standard result that a sharp geometric obstruction mixes angular-momentum channels. The barrier locally breaks $C_{6v}$ down to $C_{2v}$ along the barrier line, lifting the $C_6$ selection rule that separates $m = \pm 1$ from $m = \pm 2$. The $p$-wave $\to$ $d$-wave conversion is strong because the slit width (four lattice units) is comparable to the wavelength $\lambda \approx 21$ physical units, putting the

experiment in the Fresnel near field. Figure 5 summarises the mode content as a bar chart.

### D. Near-field regime

With $\lambda = 2\pi/k_0 \approx 20.9$ physical units, a barrier-to-screen distance $L \approx 21.7$, and a slit separation $d = 9$, the ratio $L/\lambda \approx 1$ places the experiment firmly in the Fresnel near-field regime, not the Fraunhofer far field. The Fraunhofer formula $\Delta y = \lambda L/d \approx 50$ is therefore larger than the available screen extent; what we observe on the screen is the expected Fresnel pattern with three dominant maxima located approximately at slit A, the midpoint, and slit B. This is the physically correct near-field response and confirms that the lattice reproduces ordinary Huygens–Fresnel diffraction once the physical-band projection is used.

### E. Square-lattice comparison

For reference, we have re-run the same experiment on a 2+1D *square* lattice with four lightlike moves $(\pm1, 0, \pm1)$, $(0, \pm1, \pm1)$ and the same $i\varepsilon$ amplitude rule (the natural 2+1D extension of the Feynman checkerboard [2]). The square lattice has $c = 1$ and only a 3-dimensional physical band at $k = 0$, versus the 5-dimensional band of the hexagonal lattice. A wave packet propagates with a forward-versus-sideways density asymmetry of factor $\sim$40 at modest distance—the residual anisotropy of the four-direction lattice. The hexagonal lattice uses 33% more grid points per wavelength on the $y$-axis (because $\Delta y_{\rm hex} = 0.75$ versus $\Delta y_{\rm sq} = 1$) but in exchange delivers exact $C_{6v}$ isotropy and the richer 5-fold vector-plus-$d$-wave band structure.

## V. BOX STATES AND SYMMETRY BREAKING BY CONFINEMENT

### A. Setup

In the 3+1D cuboctahedral lattice we impose hard-wall boundary conditions on a cubic box of side $N$ lattice units (amplitude is forced to zero outside the box). The half-step transfer matrix $M_{\rm half}$ is built as a sparse $26N^3 \times 26N^3$ complex CSR matrix; the full-step transfer matrix $M_{\rm full} = M_{\rm half}^2$ is assembled and passed to ARPACK in shift-invert mode, with the shift $\sigma = (1+\varepsilon^2)\exp(-2i\varepsilon)$ chosen at the bulk-band eigenvalue $\lambda_{\rm band} = 0.99 - 0.20i$. We compute the 40 eigenvalues closest to $\sigma$ at each box size $N \in \{7, 9, 11\}$ and classify each eigenvector into $O_h$ irreps using the projection operator

$$\frac{\|\mathcal{P}_\rho \psi\|^2}{\|\psi\|^2} \;=\; \frac{d_\rho}{48} \sum_{g \in O_h} \chi_\rho(g)\, \langle \psi | R(g) | \psi \rangle, \qquad (12)$$

where $R(g)$ acts jointly on the spatial site (around the box center) and on the 13 direction indices. Every box eigenvector is found to be 100% in a single irrep (residual $< 10^{-6}$), confirming that the box respects the full $O_h$ symmetry.

### B. Energy level ordering

The lowest eigenvalue in each band irrep is listed in Table III. At every box size studied, we find the same consistent ordering

$$E(T_{1u}) \;<\; E(T_{2g}) \;<\; E(E_g), \qquad (13)$$

with $T_{1u}$—the 3-dimensional vector irrep—the ground state of the band.

### C. The $1/L^2$ scaling

The gap $\Delta E \equiv E(E_g) - E(T_{1u})$ in Table III shrinks monotonically with the box side $N$. Taking the ratio of the $N = 7$ and $N = 11$ gaps gives

$$\frac{\Delta E(N=7)}{\Delta E(N=11)} \;=\; \frac{0.01756}{0.00696} \;=\; 2.52, \qquad (14)$$

which is to be compared with the prediction of a purely kinematic $1/L^2$ scaling,

$$\left(\frac{11}{7}\right)^2 \;=\; 2.47. \qquad (15)$$

TABLE I. The general exclusion law for the equilateral path-integral lattice. $N$ is the number of lightlike diagonal directions; the physical band has dimension $N-1$; the $s$-wave is excluded with $C$-eigenvalue $1 + Ni\varepsilon$.

| Dim. | $N$ | band dim. | excluded | $\lambda_C^{(s)}$ |
|---|---|---|---|---|
| 1+1D triangular | 2 | 1 | $A$ | $1+2i\varepsilon$ |
| 2+1D hexagonal | 6 | 5 | $A_1$ | $1+6i\varepsilon$ |
| 3+1D cuboctahedral | 12 | 11 | $2A_{1g}$ | $1+12i\varepsilon$ |
| general | $N$ | $N-1$ | $A_1$ | $1+Ni\varepsilon$ |

TABLE II. Fractional crystal-angular-momentum content of the physical amplitude at representative points on the 2+1D double-slit geometry ($t = 60$). "Free upstream" is the free wave ahead of the barrier; "slit center" is just past each open slit; "slit edge (past)" is immediately outside a slit opening, +2 nodes downstream of the barrier.

| point | $m=+1$ | $m=-1$ | $m=+2$ | $m=-2$ | $m=+3$ |
|---|---|---|---|---|---|
| free upstream | 0.492 | 0.492 | 0.006 | 0.006 | 0.005 |
| slit A center | 0.269 | 0.549 | 0.078 | 0.086 | 0.018 |
| slit B center | 0.549 | 0.269 | 0.086 | 0.078 | 0.018 |
| slit A edge | 0.358 | 0.092 | 0.077 | 0.397 | 0.076 |
| slit B edge | 0.092 | 0.358 | 0.397 | 0.077 | 0.076 |

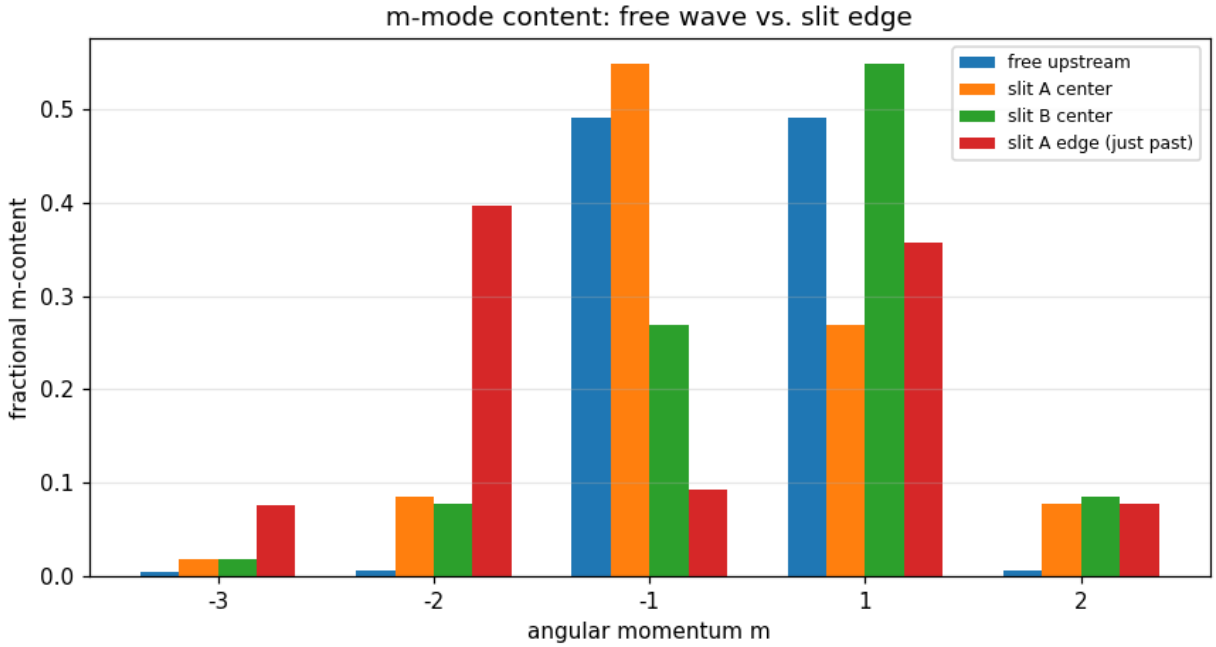


FIG. 5. Crystal-angular-momentum content of the physical amplitude at $t = 60$ in the hexagonal double-slit geometry. Free upstream: the wave is almost pure $m = \pm 1$ (vector sector). Slit centers break the left–right symmetry and pick one chirality. At slit edges, $d$-wave content ($m = \pm 2$) reaches $\sim$40%, a roughly 40–80$\times$ enhancement over the free wave.

The agreement is better than 2%. Physically the splitting is therefore purely kinematic—analogous to the $n^2/L^2$ energy levels of a particle in a box—and does not indicate any intrinsic energetic preference for one irrep over another beyond that induced by confinement.

### D. $T_{1u}$ as vector ground state

Combining Sections IV and V, the vector irrep $T_{1u}$ emerges as the natural low-energy sector of the lattice along two independent axes:

1. In *free propagation*, a smooth wave packet is 98% $T_{1u}$ in its angular-momentum content.
2. In *confinement*, $T_{1u}$ is the lowest-energy band irrep at every box size tested, and other band irreps $(T_{2g}, E_g)$ are separated from it by a gap that scales kinematically as $1/L^2$.

TABLE III. Lowest box-state energy $E = -\arg\lambda$ in each band irrep as a function of box side $N$ for the 3+1D cuboctahedral lattice at $\varepsilon = 0.1$. The bulk reference is $E = 2\varepsilon/(1-\varepsilon^2) \approx 0.199$; all finite-size values lie below it and approach it from below as $N \to \infty$. $T_{2u}$ does not appear within the 40 eigenvalues closest to the bulk band for any $N$ studied; it has a larger finite-size shift.

| $N$ | $E(T_{1u})$ | $E(T_{2g})$ | $E(E_g)$ | gap $E_g - T_{1u}$ |
|---|---|---|---|---|
| 7 | 0.05078 | 0.05201 | 0.06834 | 0.01756 |
| 9 | 0.07468 | 0.07506 | 0.08350 | 0.00882 |
| 11 | 0.08419 | 0.08601 | 0.09115 | 0.00696 |

Neither result is forced by any tuning: no mass, coupling, or boundary parameter was adjusted. The effect is purely geometric, arising from the cuboctahedron direction space and the universal $i\varepsilon$ amplitude rule.

Under the continuous rotation group SO(3) the lattice irrep $T_{1u}$ flows to the spin-1 vector representation in the long-wavelength limit $k \to 0$, the same representation carried by every gauge boson in the Standard Model (photon, $W^\pm$, $Z$, gluons). We do not claim to have derived gauge theory from this lattice; we merely note that the geometric ground state of the present model is a vector field, in a sense precisely analogous to how crystal momentum becomes continuous momentum in a long-wavelength limit.

## VI. DISCUSSION

### A. The exclusion law is universal

The spectral gap (11) depends only on the combinatorics of the $N$ diagonal directions, not on any geometric detail other than "the moves are related by a transitive group action and the amplitude rule is $\delta_{dd'} + i\varepsilon(1-\delta_{dd'})$". A square lattice with $N = 4$ lightlike directions should therefore give a 3-dimensional physical band with $s$-wave

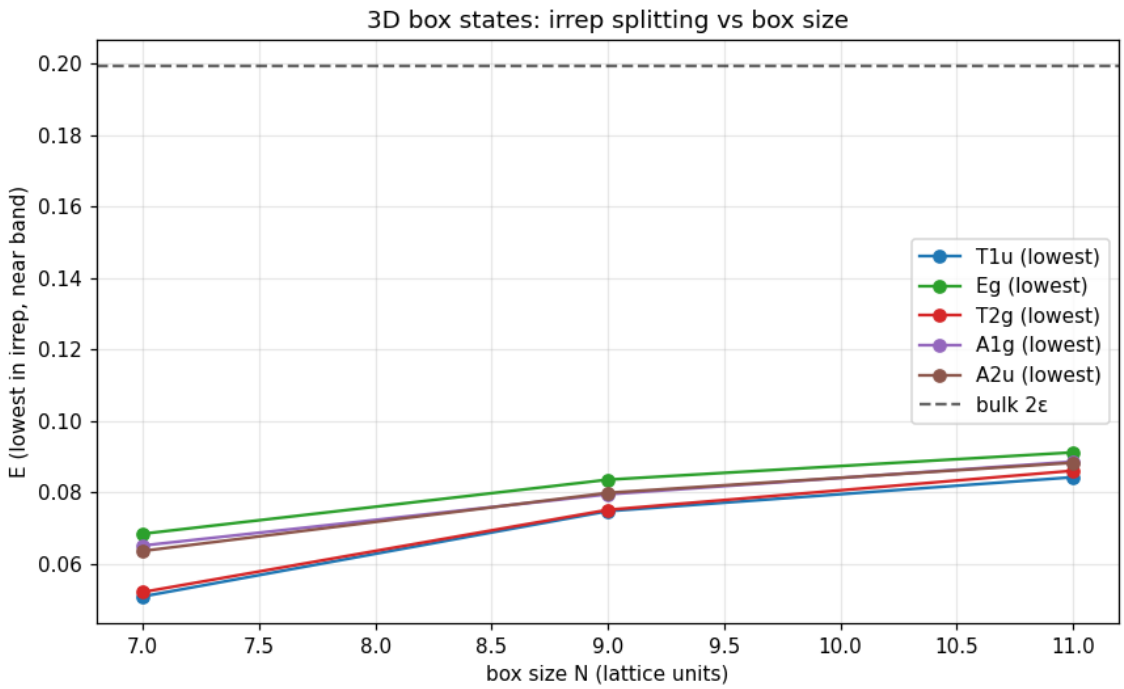


FIG. 6. Lowest energy per $O_h$ irrep as a function of box side $N$ for hard-wall box states on the 3+1D cuboctahedral lattice ($\varepsilon = 0.1$). At every box size $T_{1u}$ is the lowest of the observed band irreps, followed by $T_{2g}$ and $E_g$. The dashed black line is the bulk band value $E = 2\varepsilon \approx 0.199$. The $T_{1u}$–$E_g$ gap follows $1/L^2$ to within 2%.

eigenvalue $1+4i\varepsilon$; this is exactly what Section IV finds for the 2+1D square checkerboard. The law is a geometric fact about the direction space, not about the dimensionality of spacetime.

### B. Crystal versus continuous angular momentum

The crystal angular momenta $m \in \{\pm 1, \pm 2, +3\}$ in $C_{6v}$ and the irreps $E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}$ in $O_h$ are discrete: they label finite-group representations, not SO(2) or SO(3) irreps. Like crystal momentum in a solid-state band structure, these labels become continuous in the long-wavelength limit. In that limit $T_{1u}$ flows to a continuous spin-1 vector, $T_{2u}$ to its parity partner, and $E_g \oplus T_{2g}$ to the $\ell = 2$ ($d$-wave) sector. The $k = 0$ degeneracy of all four irreps within the physical band is therefore a genuine *lattice* signature; it is protected by $O_h$ and broken only away from $k = 0$ or by confinement.

### C. Vector ground state—implications

The combined results of Sections IV and V produce a specific and geometric statement: the lowest-energy, freely-propagating content of the equilateral lattice is a vector field. Higher angular-momentum content ($E_g$, $T_{2g}$, the $d$-wave) is not absent—it appears at finite $k$ via the little-group splitting of Fig. 3, and at obstructions via the slit-edge enhancement of Fig. 5—but it is an *excited* response of the lattice, not the ground state.

This is a suggestive geometric counterpart to the empirical fact that every fundamental interaction in nature is mediated by a spin-1 vector boson. It does not, by itself, derive gauge invariance or explain the Standard Model: it merely points out that the *kinematic* vacuum of a geometrically minimal relativistic lattice is a vector field, while scalars and tensors require extra input (obstructions, external momentum, or additional lattice structure).

### D. Open questions

The present analysis leaves four questions open. First, the $T_{2u}$ irrep lies outside the 40-eigenvalue shift-invert window in every box size studied. From the band decomposition (9) and the observed ordering (13) we predict $E(T_{2u}) > E(E_g)$, but a direct check requires a wider ARPACK window or an alternative diagonalisation. Second, the band-tracker caveat noted in Ref. [1] remains: at $k > 0$ the 11-fold band fragments into several sub-bands, and identifying the isotropic $T_{1u}$ sub-band requires a more careful eigenvector-overlap tracker than we have implemented here. Third, whether a U(1) or non-abelian gauge symmetry can emerge from couplings between $T_{1u}$ modes is an open question: the $T_{1u}$ ground state carries a 3-component "vector potential" but no gauge constraint

has been derived. Fourth, explicit $O_h$-breaking by non-cubic boxes should shift the level ordering; whether $T_{1u}$ remains the ground state under such perturbations is a natural numerical test.

## VII. CONCLUSION

We have shown that the internal structure of the equilateral path-integral lattice is fully determined by the geometry of its direction space. The physical propagating band is the traceless part of the $N$-dimensional diagonal-direction Fourier representation, the $s$-wave being excluded by a factor-$N$ spectral gap in the local amplitude matrix. In 2+1D this produces a 5-fold band $E_1 \oplus E_2 \oplus B$ carrying crystal angular momenta $m \in \{\pm 1, \pm 2, +3\}$ of $C_{6v}$. In a natural 3+1D cuboctahedral extension it produces an 11-fold band $E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}$ of the full octahedral group $O_h$.

Numerical double-slit and hard-wall box experiments on these lattices both single out the vector irrep $T_{1u}$ as the natural low-energy sector: free wave packets are 98% $T_{1u}$, and the box ground state is $T_{1u}$ at every box size studied, with a $T_{1u}$–$E_g$ gap that scales as $1/L^2$ to within 2%. Higher-angular-momentum content is only excited at obstructions or at finite momentum.

Geometry → relativistic dispersion [1] → internal structure → vector ground state: the chain from lattice edges to an emergent vector field is now explicit. The natural next step is to study interactions and couplings between $T_{1u}$ modes, and to ask whether gauge invariance can be recovered as a long-wavelength consequence of the lattice geometry alone.

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